

2024 AMC 10A Problem 18 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10A Problem 18. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10A Problem 18

Problem:

There are exactly K positive integers b with $5 \leq b \leq 2024$ such that the base- b integer 2024_b is divisible by 16 (where 16 is in base ten). What is the sum of the digits of K ?

Answer Choices:

- A. 16
- B. 17
- C. 18
- D. 20
- E. 21

Solution:

Notice that $2024_b = 2b^3 + 2b + 4 = 2(b+1)(b^2 - b + 2)$, and consider the residue classes of this number modulo 8. If $b \equiv 7 \pmod{8}$, then $b+1 \equiv 0 \pmod{8}$, and if $b \equiv 3$ or $6 \pmod{8}$, then $b^2 - b + 2 \equiv 0 \pmod{8}$. In each case 2024_b is divisible by 16.

In all other cases 2024_b is not divisible by 16. Indeed, if $b \equiv 0, 2$, or $4 \pmod{8}$, then $b+1$ is odd, and $b^2 - b + 2 \equiv b+2 \pmod{8}$, so 2024_b is divisible by no power of 2 greater than 2^3 . If $b \equiv 1$ or $5 \pmod{8}$, then $b+1$ and $b^2 - b + 2$ are both odd multiples of 2, so 2024_b is divisible by 8, but not by 16.

Because $2024 = 253 \cdot 8$, there are $253 \cdot 3 = 759$ positive integers $b \leq 2024$ that are congruent to 3, 6, or 7 modulo 8. The number 3 must be

excluded from this total, because the problem statement requires b to be at least 5. Thus $K = 759 - 1 = 758$, and the sum of the digits of K is $7 + 5 + 8 = (E)\boxed{20}$.

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